

Assignment 1: Text Cleaning & Tokenization (1 Hour)

- Take a small dataset of 20–30 text reviews (movie/product reviews).
- Perform:
 - * Lowercasing
 - * Removing punctuation
 - * Removing stopwords
 - * Tokenization using NLTK or spaCy
- Submit cleaned tokens output and explanation of each preprocessing step.

Assignment 2: TF-IDF Implementation (1 Hour)

- Use sklearn TfidfVectorizer on the same dataset.
- Convert text into TF-IDF vectors.
- Display:
 - * Feature names
 - * Shape of matrix
 - * Top 10 important words per document
- Write short explanation on:
 - * What high TF-IDF value means?
 - * Why IDF is important?

Assignment 3: Word2Vec Embeddings (1 Hour)

- Train a simple Word2Vec model using gensim on your dataset.
- Find:
 - * Similar words to a chosen word (e.g., “good”)
 - * Vector size and shape
- Compare:
 - * Difference between TF-IDF and Word2Vec in 5–6 bullet points.
- Bonus: Try using a pre-trained model (Google News vectors if possible).

Submission Requirements:

- Jupyter Notebook with proper comments
- Short PDF summary (1–2 pages) explaining learnings
- GitHub upload with README